

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (EE) Sem - III Examination (Backlog), 2011
EE - 207 Network Analysis

Date: 17.05.2011, Tuesday Time: 10:00a.m to 1:00p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary.

SECTION - I

- Q-1 A. State and explain Superposition theorem. Find Current I in the circuit shown in Fig.1 5
by superposition theorem.

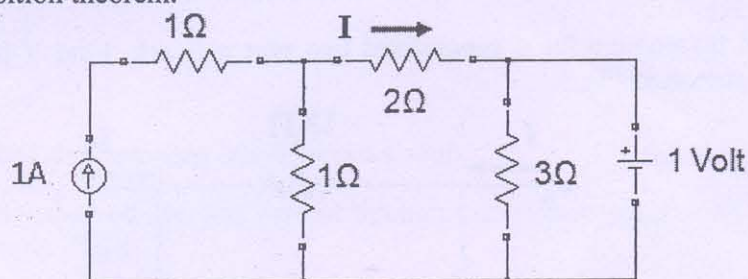


Fig. 1

- B. Determine the power loss in $1\ \Omega$ resistor for circuit shown in fig.2 by Thevenin's 5
theorem.

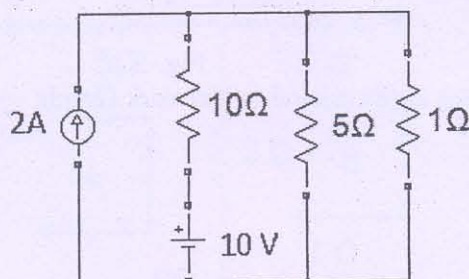


Fig.2

- Q-2 A. State reciprocity theorem. Show the application of reciprocity theorem to network 5
shown in Fig.3.

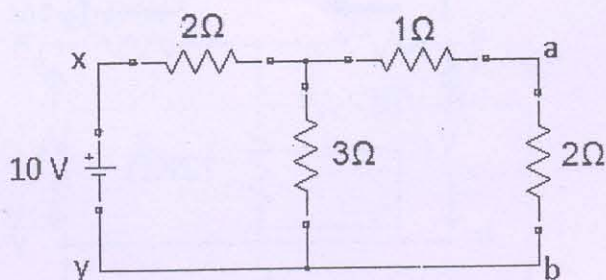


Fig.3

B Find ABCD parameters in terms of Y parameters for generalized two port network 5

OR

Q-2 A. Find the current through $1\ \Omega$ resistor for the circuit shown in Fig. 4 using Norton's theorem. 5

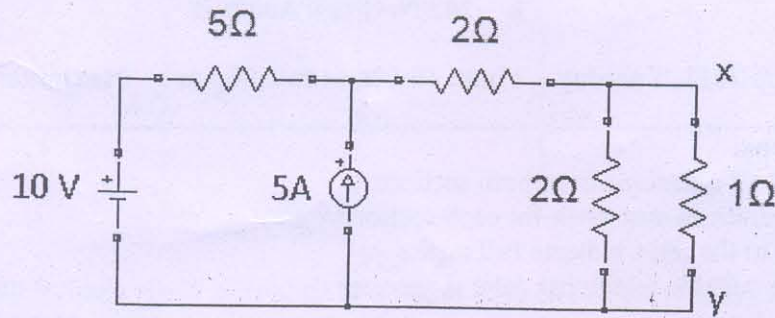


Fig. 4

B. Find Hybrid parameters for a generalized two port network 5

Q-3 A. Derive Z parameters for a generalized two port network. Find Y-parameters for the network shown in Fig.5. 7

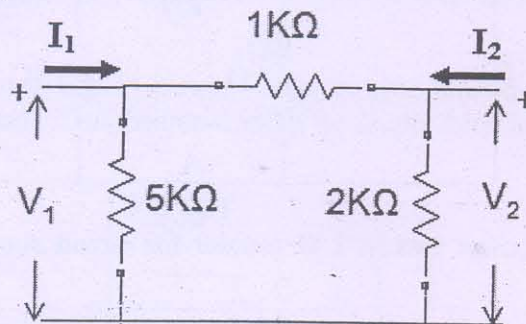


Fig. 5

B. Explain the following terms related to Network Graph:

- Node
- Tree
- Tie-set
- Branch

8

OR

Q-3 A. Derive Y parameters for a generalized two port network Find Z-parameters for the network shown in Fig.6 8

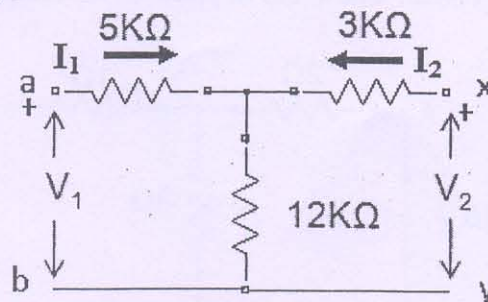


Fig. 6

- B. What is Cut-set matrix. Show the cut-sets for the graph of the network shown in Fig. 7 and develop the fundamental cut-set matrix.

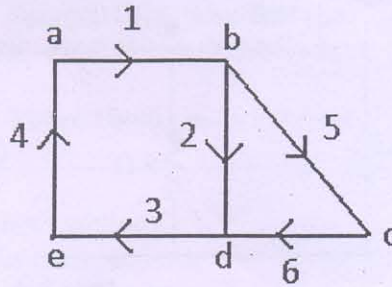


Fig. 7

SECTION - II

- Q-4 A. Obtain laplace transform of following functions: 6
- $e^{-\alpha t}$
 - $\sin(\omega t)$
 - t^2

- B. Explain in detail step response of R-C series circuit. 4

- Q-5 A. Determine the node voltages and current through the resistor using mesh method for network shown in Fig. 8 5

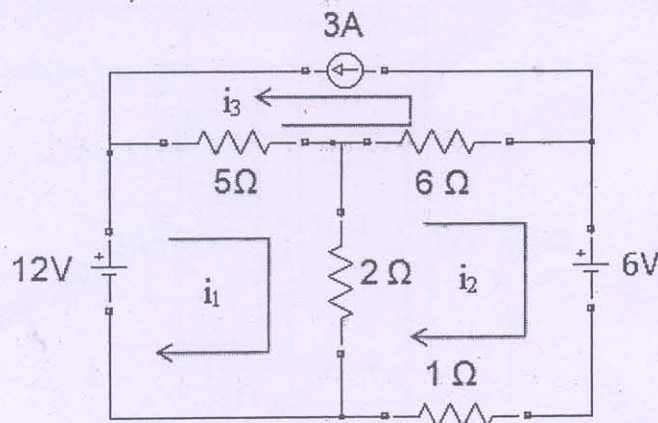


Fig. 8

- B. For the circuit shown in Fig. 9, $V = 10\text{ V}$, $R = 10\ \Omega$, $L = 1\text{H}$, $C = 10\ \mu\text{F}$ and $V_c(0) = 0$. Find $i(0+)$, $di/dt(0+)$ and $d^2i/dt^2(0+)$. 5

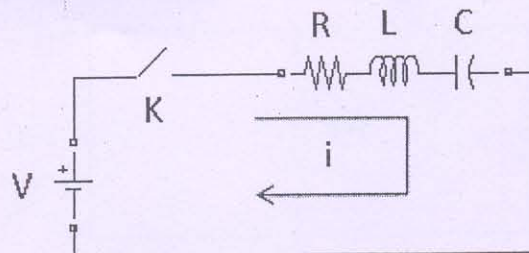


Fig. 9

OR

- Q-5 A. Find the current through 5Ω resistor shown in Fig. 10 by Nodal analysis

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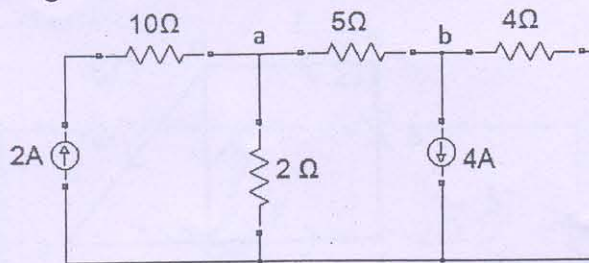


Fig. 10

- B. A unit ramp voltage $r(t-3)$ is applied to a series R-L circuit [$R = 5\Omega$; $L = 2H$] at $t=0$. Find its response with respect to current. [Assume zero initial condition]

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- Q-6 A. A unit impulse function of voltage is applied at $t=0$ to a series R-C circuit. Find $i(t)$ assuming the initial charge stored in the capacitor to be zero. For the same circuit if the excitation voltage is now a shifted impulse function $\delta(t-a)$ and assuming zero initial conditions, find $i(t)$. Assume $R = 5\Omega$, $C = 2F$.
B. Draw and write expression for different basic types of special signals.

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OR

- Q-6 A. In the network shown in Fig. 11 a steady state is reached with switch K open, and at $t=0$ the switch is closed. Find the initial value for all the three loop currents.

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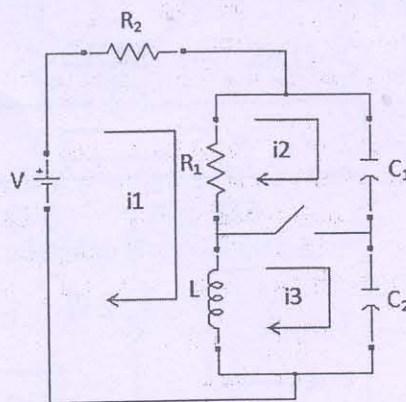


Fig. 11

- B. Derive laplace transformation of basic types of special signal waveforms.

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